Radiometric Control Witness Form & Diagrams

Project Name: Lenton Lodge Gatehouse, Derby Road, Lenton, Nottingham, Nottinghamshire, NG7 2QA, UK **Date:** 15/01/2024
Project Code: GATE

	Details	Notes/Comments	Justification/Explanation
Survey Overview			
Date & Time:	15/01/2024 10:40 (arrival) 12:00 (leave)	Date and time of survey. Time is important	Same day as the thermal imaging survey.
General Survey Conditions:	Temp: -1°C WB: 0–1°C DP: -7.2°C WS: 0.5m/s – 2.0m/s	Brief description of survey conditions (inc., live and recent weather, visibility, cloud cover, air quality.	No rain for 24hrs, cold (-3-3°C) Sunny, clear day, frost on ground, cloudless, cold
No. of Points	2	Number of points being taken. Distribution of control and check points	
Equipment:			
Make / Model:	RS PRO RS-8876 InfraRed Thermometer	Make/model of radiometric equipment used.	
Specifications:		Critical specifications only (inc., temp. range, accuracy, sensitivity, recorded metrics, spot ratio)	
Additional Equipment		Any additional equipment needed for the execution of the survey.	
Survey Team:			
Planned / Set Up by:	Neil Sutherland	Lead individual responsible for survey planning	
Observed by:	Neil Sutherland	The individual taking the point observations. This should remain consistent during the survey.	

Computed by:	Neil Sutherland	The team member responsible for the processing of the observations	
Environmental Conditions:			
Air Temperature (AT)	3.8-4.4°C		Details of recording instruments included in RadiometricControl.md
Reflected Temperature (RT)	1.1°C		Details of recording instruments included in RadiometricControl.md
Relative Humidity (RH)	41-50%		Details of recording instruments included in RadiometricControl.md
Emissivity:			
Observed Materials & Emissivities	Mansfield Sandstone (0.70)	Listed the materials observed and their emissivities. This includes any values determined outside of this survey.	
Observation Method	TESTO Emission tape	Brief description of how emissivity was observed and calculated (inc., no. of observations and redundancy)	
Notes:			
General Notes:			

POINT 1

	Details	Notes/Comments	Justification/Explanation
Point Name:	**P1_Rubber**		
Temperature (°C):	1.3°C	Measurement taken at ~1m from target, where red laser was covering black rubber element	
Emissivity (ε):	0.91	Previously determined using ASTM E1933-14 method	
Material:	Rubber		

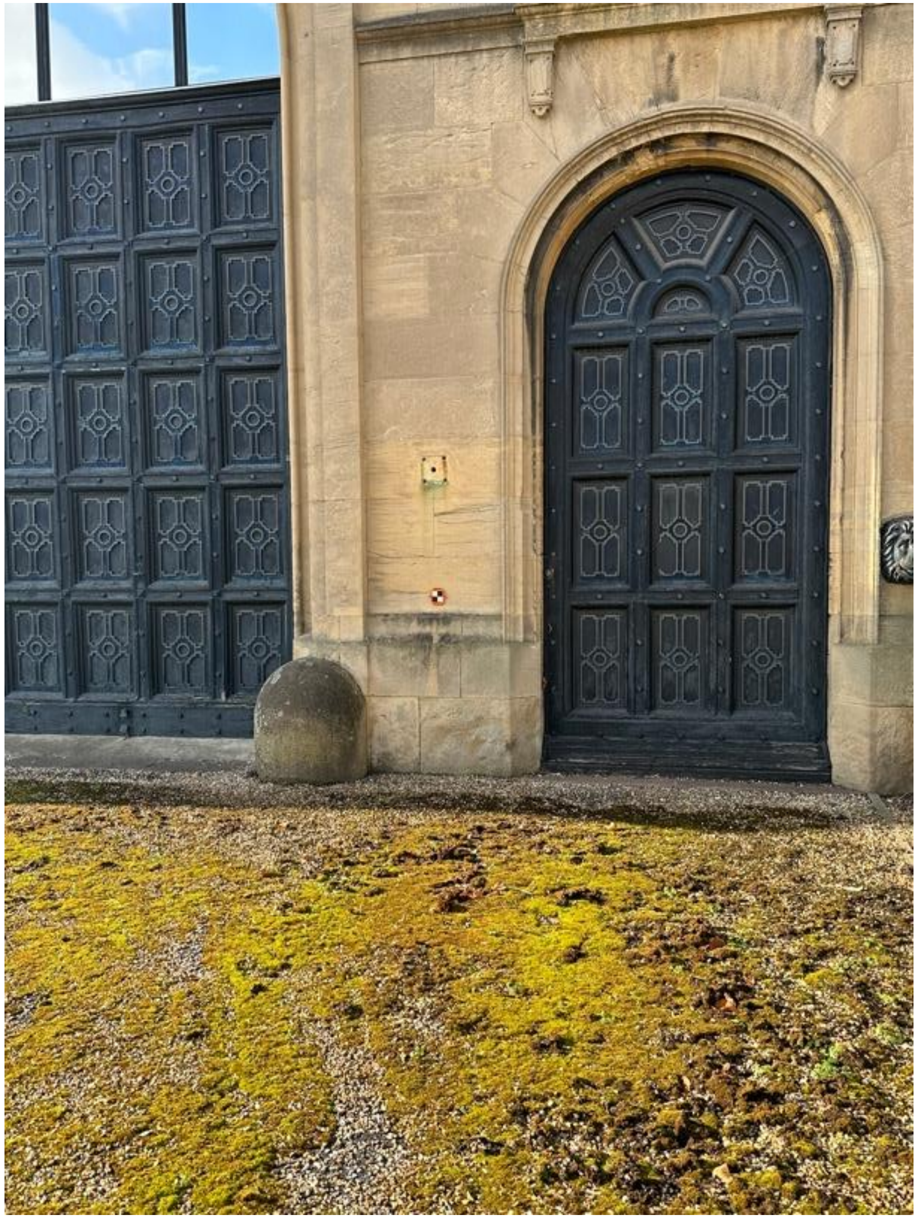
Notes:	
Notes:	Measurement taken on TOP LEFT black rubber on thermal-specific survey marker.

Diagrams	
Control Point Photo(s):	

POINT 2

	Details	Notes/Comments	Justification/Explanation
Point Name:	**P5_Rubber**		**Point Name:**
Temperature (°C):	1.9°C	Measurement taken at ~1m from target, where red laser was covering black rubber element	**Temperature (°C):**
Emissivity (ε):	0.91	Previously determined using ASTM E1933-14 method	**Emissivity (ε):**
Material:	Rubber		**Material:**

Notes:	
Notes:	Measurement taken on TOP LEFT black rubber on thermal-specific survey marker. This side received greater solar radiation during the survey, indicated by the slightly increased temperature compared to P1_rubber

Diagrams	
Control Point Photo(s):	

Additional Images / Diagrams

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